

1 STACKING — NET SF BY LEVEL (BASE, FULL PROGRAM, 50/50 SEATS)

ROOM / SPACE	NET SF	WHY THIS LEVEL	SF STATUS
LEVEL 1 (GROUND) — MUST STAY AT GRADE	45,713		
Championship Wrestling Arena (event floor)	22,000	double height; nothing above	TAG · HIGH
Lower seating tier (1,100 seats)	6,600	lower tier, inside arena volume	CITED · MED
Boys Locker Room	3,500	next to the floor (DR1)	TAG · HIGH
Girls Locker Room (as tagged; D-013)	3,503	next to the floor (DR1; D-013)	TAG · HIGH
Event / visitor locker rooms (DR3)	3,600	under bowl, next to floor (DR3)	ASSUMED · LOW
Foyer / trophy lobby (hall of champions)	800	main entry at grade (DR5)	CITED · MED
Lower concourse	1,375	25% of lower seats x 5 SF	ASSUMED · LOW
Public restrooms L1 (41 fixtures)	2,050	T2902.1 on L1 load x 50 SF	CITED · MED
Concession stand	275	main concourse (ASSUMED)	CITED · LOW
First aid	120	at grade for EMS (ASSUMED)	CITED · MED
Equipment storage (mats + floor covers)	700	mats roll onto floor (ASSUMED)	CITED · LOW
Equipment room	300	service at grade (ASSUMED)	CITED · LOW
Stairs (4) + elevator (1)	890	stairs + elevator (both levels)	CITED/ASSUMED
Mechanical (5% of total gross)	4,433	at grade (ASSUMED)	CITED · LOW

LEVEL 2 / MEZZANINE — CAN GO UP	20,785		
Upper seating tier (1,100 seats)	6,600	upper tier over the L1 ring	CITED · MED
Strength & Conditioning	6,000	daily room; gym floor loads	TAG · HIGH
Cross-Training Mat Area	4,000	daily mat IS the mezzanine (DR4)	TAG · HIGH
Administrative office	470	no grade need; desk in foyer	CITED · LOW
Upper concourse + champions wall	1,375	25% of upper seats x 5 SF	ASSUMED · LOW
Public restrooms L2 (29 fixtures)	1,450	T2902.1 on L2 seats x 50 SF	CITED · MED
Stairs (4) + elevator (1)	890	stairs + elevator (both levels)	CITED/ASSUMED

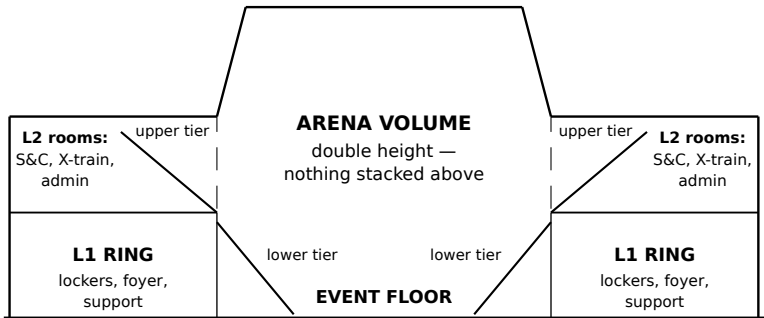
Bold rows = ARENA VOLUME (double height). Stage + utility room: no source, left out (TBD).

L1 GROSS = 1.25 x (L1 net + mechanical)	62,683
of which ARENA VOLUME = 1.25 x (event floor + lower tier)	35,750
of which L1 RING (rooms that can carry L2 above)	26,933
L2 GROSS = 1.25 x L2 net (must fit over the L1 ring)	25,981
GROUND FOOTPRINT = max(L1, arena volume + L2) vs 55,000 cap	62,683
TOTAL GSF, BOTH LEVELS (no longer capped, D-031)	88,664

Same factors as Rev A (frozen): gross-up 1.25 base / 1.15 lean (R-014), seats 6.0 SF base / 3.38 lean (R-008), mechanical 5% of gross. Each level is grossed up on its own. Rev A (one level) needed ≈ 86,100 GSF base.

Not credited (conservative): rooms under the lower seating tier, upper-tier overhang past the L1 ring, rooftop mechanical. Lean case: lower tier is telescopic, so nothing can be built under it.

2 BOWL GEOMETRY — SECTION (NTS)



GROUND FOOTPRINT = arena volume + L1 ring

The arena's high volume stays open to the roof, so no rooms go over the event floor or lower tier. The upper tier and every L2 room sit on the ring of L1 rooms around it (rooms under the seating). Fit needs L2 ≤ L1 ring; otherwise the ring grows and so does the footprint.

3 VERTICAL CIRCULATION — ON BOTH LEVELS

L2 occupant load (base, 50/50) = 1,100 seats + 204 in S&C, cross-training, admin (T1004.5) = 1,304 -> 4 exits (T1006.3.3). Stair capacity 0.2 in/occupant (1005.3.1 exc.: sprinklers + voice alarm, see below) -> 261 in total = 4 stairs x 65 in (min 44 in, 1011.2; any 1 lost keeps ≥ 50%, 1005.5).

Each stair: 15 ft floor-to-floor (ASSUMED) = 26 risers of 6.92 in (≤ 7 in) and 11 in treads (1011.5.2), 2 flights ≤ 12 ft rise (1011.8), landings 48 in deep (1011.6): ≈ 206 SF per level. Elevator: 1 (IBC 1104.4 accessible route to a story over 3,000 SF), car ≥ 80 x 54 in (ADA T407.4.1), hoistway 64 SF per level (ASSUMED). Total ≈ 890 net SF on EACH level.

Open stairs OK between only two stories (1019.3 exc.). Sprinklers required: A-4 fire area > 12,000 SF or 300+ occupants (903.2.1.4); voice/alarm at 1,000+ (907.2.1.1). Sprinklered, the exit stairs serve as accessible means of egress without areas of refuge (1009.1, 1009.3.2, 1009.3.3); elevator egress only at 4+ stories (1009.2.1).

4 RESTROOMS PER LEVEL (T2902.1)

L1 load 1,540 (lower seats + event floor ÷ 50): WC 11 M / 20 W, lav 4 / 6 = 41 fixtures. L2 load 1,100 seats: WC 8 / 14, lav 3 / 4 = 29. Sized per level (conservative; 2902.3.3 allows one story of travel). 50 SF/fixture ASSUMED. Rev A one-level count: 67.

5 RESULT — FOOTPRINT VS 55,000 CAP

FULL PROGRAM, 2 LEVELS:

BASE ≈ 62,300 SF — OVER by ≈ 7,300

LEAN ≈ 51,600 SF — FITS (3,400 spare)

Total GSF ≈ 88,900 base / ≈ 74,700 lean (best seat split)

CASE	SPLIT	FOOTPRINT	TOTAL GSF	FIT?
Base, 22,000 floor	50% up (assum.)	62,683	88,664	NO +7,700
Base, 22,000 floor	55% up (best)	62,303	88,916	NO +7,300
Base, 14,364 floor (B)	50% up (assum.)	52,235	78,216	YES
Lean, 22,000 floor	50% up (assum.)	53,518	74,107	YES
Lean, 22,000 floor	65% up (best)	51,631	74,678	YES
Lean, 14,364 floor (B)	50% up (assum.)	43,957	64,546	YES
Lean, 14,364 floor (B)	60% up (best)	42,559	64,825	YES

Full program = 22,000 SF floor + 2,200 seats + all 5 tags. Split = share of seats in the upper tier (50/50 ASSUMED; 'best' = lowest footprint in 30-70%). Sightlines not checked.

6 SMALLEST CHANGE THAT FITS (BASE)

Option B floor (14,364 SF, 4 mats 42 ft + tables + 1 court): footprint ≈ 52,200 — fits with all 2,200 seats; total ≈ 78,200 GSF.

Or keep the floor and cut to ≈ 1,370 seats (–830; best split). Any floor ≤ ≈ 16,400 SF fits with all seats.

Or build to the lean inputs (telescopic lower tier, 1.15 gross-up): full program fits as is.

7 CITED vs ASSUMED

CITED: code rules (IBC 2021 ch. 10, 11, 9; ADA 2010), seat and gross-up factors, fixture ratios, lockers-at-event-level precedent (Reed, Orleans arenas). ASSUMED (verify): 15 ft floor-to-floor, 64 SF hoistway, 50/50 split, which rooms go up, mechanical at grade, 50 SF/fixture, concourse 25%, A-4 occupancy.

8 SOURCES (retrieved 2026-10-03)

- IBC 2021 (UpCodes, AL ed.): 903.2.1.4, 907.2.1.1, 1005.3.1, 1005.5, T1006.3.3, 1009, 1011.2/.5.2/.6/.8, 1019.3, 1104.4, 2902.3.3 — R-015
- 2010 ADA Standards (access-board.gov) 206.2.3, T407.4.1 — R-015
- Reed Arena Facility Guide; Orleans Arena Production Guide 2024 (venuecoalition.com) — R-015
- Rev A sources: R-008, R-009, R-012, R-014, R-005 (see Rev A)

PRELIMINARY — CONCEPTUAL DESIGN — NOT FOR CONSTRUCTION

PROJECT	PHASE	SHEET TITLE	SCALE	DATE	REV	DRAWN BY	SHEET
Hazel Green Regional Athletic Complex Trojan Horse Arena · Hazel Green area, Madison County, Alabama (site TBD, D-006)	PHASE 2	PROGRAM TEST-FIT, 2 LEVELS FOOTPRINT MATH, NOT A PLAN	NTS	October 2026	B	Drawn by KEYSTONE (AI) for Shane Brazelton	P2-G-003